

CSPRNG Randomness Audit — NIST SP 800-22, all 15 tests

Source: data/csprng_dataset.bin | 100 sequences per test (63 for Random Excursions) | 188 test rows (148 are NonOverlappingTemplate sub-tests)

14/15

tests pass

0.0062

lowest uniformity p (fail < 0.0001)

94%

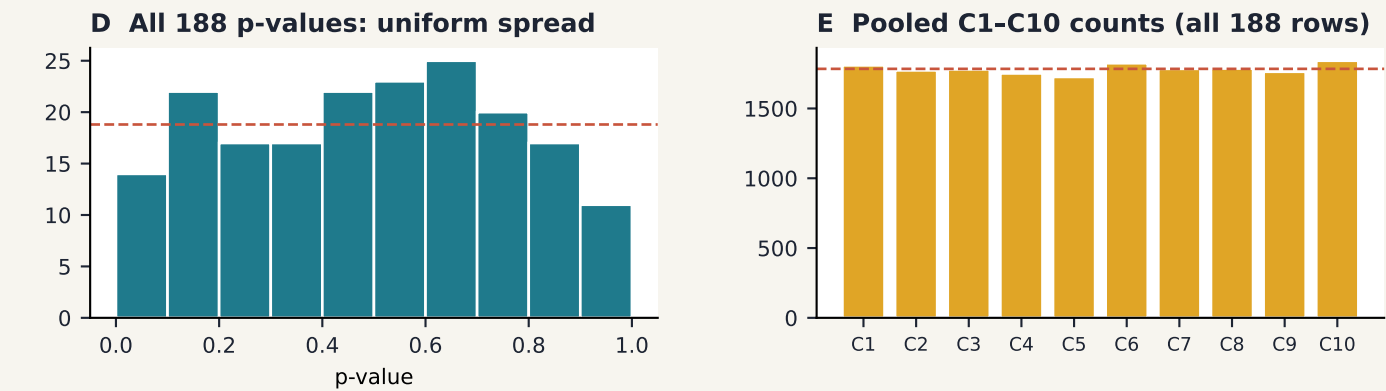
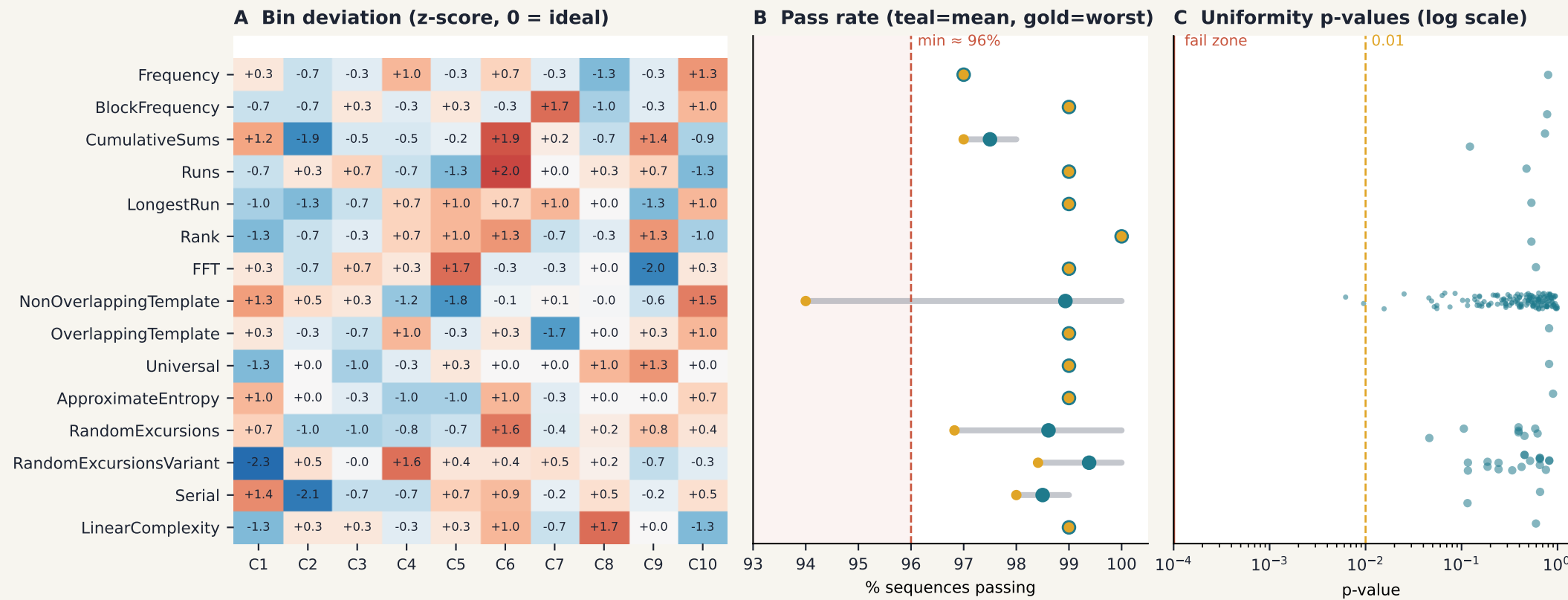
lowest pass rate (min ≈ 96%)

98.9%

mean pass rate

2

p-values < 0.01 (of 188)



Verdict: FAIL — 1 of 15 tests flagged by NIST criteria.

- Lowest pass rate 94% (min ≈ 96%); 2 of 188 rows below minimum.
- p-values < 0.0001: 0 of 188 rows.
- Pooled bins chi-square = 6.0 (9 dof; crit. 16.9).
- Flagged: NonOverlappingTemplate (2/148 rows)
- Sub-tests: NonOverlappingTemplate 148, Random Excursions (Variant) 8 (18), min 60/63.